JIANG Zixing

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EDUCATION

M.Phil. Surgery, The Chinese University of Hong Kong (CUHK), In progress 2024–

B.Eng. Electronic Information Engineering, *First-Class Honours*, CUHK-Shenzhen, 2023

EXPERIENCE

2023–24 CUHK, Faculty of Medicine
Research Assistant in the Department of Surgery
Hong Kong, China

2020–23 CUHK-Shenzhen, Robotics & AI Lab
Undergraduate Research Assistant in the Advanced Marine Robotics Group
Shenzhen, China

RESEARCH INTERESTS

Medical and Surgical Robotics

Computer-Integrated Interventional Medicine (CIIM)

PROJECTS

Autonomous Robotic Lung Ultrasonography

Research project participated at CUHK, 2023–24, supervised by Prof. LI Zheng. Project aim: use robots to perform autonomous point-of-care lung ultrasonography in intensive care units to prevent clinicians from infection during pandemic. My contribution: served as robot development lead, prototyped an robotic lung ultrasonography system and assisted in its preclinical validation.

Manipulator-Assisted Drone Landing on Dirsturbed Aquatic Platform

Research project participated at CUHK-Shenzhen, 2021–23, supervised by Prof. QIAN Huihuan. Project aim: use a manipulators to assist the landing of drones on aquatic platforms in windy and choppy conditions. My contribution: assisted Ph.D. students in the development and validation of manipulators' end-effectors and motion planning algorithms for the drone landing assistance task.

Other Selected Projects

Control of The Multi-joint Manipulator for Grasping on Water Surface – *Final Year Project, CUHK-Shenzhen*

Adaptive Control of Self-Balancing Scooter with ϵ-Greedy Q-Learning – *Course Project, Team, CUHK-Shenzhen*

Object Sorting with Eye-in-Hand Manipulator – *Course Project, Individual, CUHK-Shenzhen*

Image and LiDAR Guided Navigation for Mobile Robot – *Course Project, Individual, CUHK-Shenzhen*

Bio-Inspired Aquatic-Aerial Hybrid Locomotion Robot – *CUHK-Shenzhen Undergraduate Research Award Project*

PUBLICATIONS

Notations: * corresponding authors, # co-first authors.
Conference presenting author *italicized*.

Manuscript

2024 R. Xu, **Z. Jiang**, B. Liu, Y. Wang, and H. Qian*, "Confidence-Aware Object Capture for a Manipulator Subject to Floating-Base Disturbances," submitted to *IEEE Transactions on Robotics*, 2024 (under review).

Conference Proceedings

2023 *Y. Jiang*, R. Xu, **Z. Jiang** and H. Qian*, "Design, Modeling and Control of A Novel USV-Manipulator System," *2023 IEEE International Conference on Real-time Computing and Robotics (RCAR)*, Datong, China, 2023, pp. 206-211, doi: 10.1109/RCAR58764.2023.10249802.

2022 *C. Liu*, **Z. Jiang**, R. Xu, X. Ji, L. Zhang and H. Qian*, "Design and Optimization of a Magnetic Catcher for UAV Landing on Disturbed Aquatic Surface Platforms," *2022 International Conference on Robotics and Automation (ICRA)*, Philadelphia, PA, USA, 2022, pp. 1162-1168, doi: 10.1109/ICRA46639.2022.9812270.

Patents

2023 **Z. Jiang**, X. Ji, C. Liu, and H. Qian,"Four-wing flapping wing micro water surface aircraft and flight method," Chinese patent CN114889821B, granted February 24, 2023.

2022 X. Ji, Z. Song, **Z. Jiang**, and H. Qian, "Flapping wing mechanism and miniature water surface flapping wing aircraft," Chinese patent CN217320745U, granted August 30, 2022.

2022 X. Ji, Z. Song, **Z. Jiang**, and H. Qian, "Flapping wing mechanism based on double cranks and micro water surface flapping wing aircraft," Chinese patent CN217320744U, granted August 30, 2022.

2022 C. Liu, Z. Cao, **Z. Jiang**, R. Xu, X. Ji, and H. Qian, "Landing system, landing method and storage medium for unmanned aerial vehicle," Chinese patent CN115167522A, published October 11, 2022, patent pending.

WORKSHOP & SEMINAR ACTIVITY

2024 Presentation "A Collaborative Robotic System with In-Plane Orientation Adjustment for Lung Ultrasonograph" at workshop *Autonomy in Robotic Surgery: State of the art, technical and regulatory challenges for clinical application*, at 2024 IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, May 13, 2024.

SERVICE

Academic Conference Peer Review

The IEEE International Conference on Robotics and Biomimetics (ROBIO), 2023

LEADERSHIP

2020–22 President of Student Robotics Association, CUHK-Shenzhen

AWARDS

2023 School of Science and Engineering AY 2022–23 Dean's List Award, CUHK-Shenzhen

2021–22 The 17–19th rounds of Undergraduate Research Award, CUHK-Shenzhen

TECHNICAL SKILLS

Programming Languages: Python, C++/C, MATLAB

Robot Development: Full-stack skills with a particular emphasis on robot programming

Software: ROS, Gazebo, CoppeliaSim, SolidWorks, Docker, OpenCV, Open3D, 3D Slicer, LaTeX

Platforms: Linux, Arduino, Raspberry Pi, STM32, FPGA

LANGUAGES

Chinese	Mandarin – Native
English	Professional proficiency

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